DATA BANK

Ranking Soccer Teams

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◆ INTRODUCTION ◆

A RECENT issue of Data Bank looked at football league results as a source of statistical data. This article takes up the theme and considers alternative methods of ranking soccer teams other than by the traditional method of three competition points for a win, one for a draw and none for a loss. The criteria used involve the total goals scored and conceded by each team with regression models being used to fit prediction equations. Students can investigate techniques which may be used to determine whether some teams might have deserved to finish higher or lower than they actually did in the final standings or to be promoted or relegated. The teams considered were those in the English Premier Division during the 1992/93 season. The Premier Division results for 1993/94 are given as Table 3 at the end so that students may attempt a similar analysis for themselves.

◆ 1992/93 ANALYSIS ◆

The Premier Division of English soccer consists of 22 teams each of which plays the others twice on a home and away basis so that each team plays a total of 42 matches. The 1992/93 season record for each team is shown in Table 1. Football data are readily available from a number of media sources such as newspapers and magazines as well as books such as the annual Official Association Year Books. The rankings are decided by awarding three points for a win, one point for a draw and none for a loss. The three points for a win was changed from two points in the 1980/81 season with the idea to encourage more teams to try for a win and hence to have more goals scored. This aspect was considered by Croucher (1984) and Pollard (1986) who both reached the same conclusion that this objective was largely not met. However, the ultimate aim for any team is to finish the season with as many competition points as possible. Since home ground advantage is well documented (Edwards, 1979, Pollard, 1986, Schwartz and Barsky, 1977), some teams could endeavour to at least draw their away games and hope to win their home games. In the Premier Division this strategy would yield 84 points - the number scored by the top team Manchester United. (In Divisions One and Two it would yield 92 points - not quite enough to finish on top, but earn second place.) On the other hand, drawing **every** game would have resulted in relegation for that team from its division.

Table 1. Premier division 1992/93: win/loss/draw record, goals scored for/against and final competition points.

Team	Win	Draw	Loss	Goals for	Goals against	Final points
Man. United	24	12	6	67	31	84
Aston Villa	21	11	10	57	40	74
Norwich	21	9	12	61	65	72
Blackburn	20	11	11	68	46	71
QPR	17	12	13	63	55	63
Liverpool	16	11	15	62	55	59
Sheff. Wed.	15	14	13	55	51	59
Tottenham	16	11	15	60	66	59
Man. City	15	12	15	56	51	57
Arsenal	15	11	16	40	38	56
Chelsea	14	14	14	51	54	56
Wimbledon	14	12	16	56	55	54
Everton	15	8	19	53	55	53
Sheff. United	14	10	18	54	53	52
Coventry	13	13	16	52	57	52
Ipswich	12	16	14	50	55	52
Leeds	12	15	15	57	62	51
Southampton	13	11	18	54	61	50
Oldham	13	10	19	63	74	49
Crystal Palace	11	16	15	48	61	49
Middlesbrough	11	11	20	54	75	44
Nottm. Forest	10	10	22	41	62	40

The current criterion for ranking teams is simply to use the number of competition points accumulated by the end of the season. If you win a game by one goal or ten goals then you still only gain three competition points. Similarly, there is no extra penalty for losing by ten goals than by one goal. If teams are tied on points then the ranking criterion goes to goal **difference** i.e.

goals scored minus goals conceded. Apart from breaking ties, goals for and against seem to be of little consequence.

Since all teams play each other twice, other criteria to rank teams might include using the number of goals scored and conceded during the whole season, rather than using win/loss records. This would be similar to other sports such as, say, golf, where in major tournaments it's the final tally of strokes which counts - not how well a player did at individual holes. The question is whether considering attack and defensive capabilities over the whole season really makes any difference to the ranking system which now only considers the number of wins, draws and losses. There are a number of possibilities which might be considered to answer this question including the following ranking criteria:

(a) the **number** of goals scored for and against

(b) the goal **difference** i.e. (goals for) - (goals against)

(c) the goal **average** i.e. (goals for)/(goals against)

In each criterion a regression model is used to fit the competition points against the relevant variables. In (a) a multiple regression model is used and in (b) and (c) a simple model is used.

In each model special note of teams which were 'outliers' was taken - that is, teams whose points values were vastly different from the others based on the criterion used. The judgement used was that if a team's predicted points had a standardised residual greater than 2.0 it was omitted for purposes of constructing the model. It was only necessary to eliminate one team, Norwich.

Various transformations were also tried on the data in an attempt to improve the fit. It was found that the only transformations to provide a significant improvement was to use the **square-root** of the goal average rather than the goal average itself. All models were checked for normality and constant variance of residuals.

(a) Goals for and against

The clear outlier here was Norwich who finished with 72 competition points (3rd on the table) but whose record was 61 goals for and 65 goals against. This kind of record results from narrow wins and large losses. This was such an anomaly that removing Norwich increased the regression value of R^2 from 80.6% to 92.4%. It was also noted that Arsenal was influential since it had a small goals for (40), the lowest in the competition, combined with a low goals against (38).

The resulting model was:

$$\text{Points} = 50.4 + 0.794(\text{goals for}) - 0.687(\text{goals against}) \quad (1)$$

(b) Goal difference

Once again the outlier was Norwich which when removed increased R^2 from 78.2% to 92.0%. Manchester United was influential in the model since its goal difference of 36 was way ahead of even the second best team in this respect (Blackburn) which had a difference of 22 goals.

The resulting model was:

$$\text{Points} = 56.2 + 0.722(\text{goal difference}) \quad (2)$$

(c) Goal average

Norwich was again unusual with Manchester United being influential having a large goal for/against ratio of 2.16, the next highest being Blackburn with 1.47 followed by Aston Villa with 1.43. No other team was above 1.15.

The resulting model was:

$$\text{Points} = -13.1 + 68.377\sqrt{(\text{goal average})} \quad (3)$$

◆ SUMMARY ◆

The full summary of actual competition points and those predicted by the three models (to the nearest point) are shown in Table 2.

For the most part the predicted values are similar for each team with some notable exceptions. Norwich could perhaps be considered lucky to finish where they did in the table. By our three criteria they should have finished with around 53 points which would place them in the middle of the division instead of third. Tottenham was also lucky to finish as high as equal sixth since they conceded 66 goals - four more than Nottingham Forest who finished last. Moreover, their goals for was only 60,

Table 2. Premier division 1992/93: actual and predicted points.

Team	Actual points	Predicted competition points		
		Goals for & against (1)	Goal difference (2)	Goal average (3)
Man. United	84	82	82	87
Aston Villa	74	68	69	69
Norwich	72	54	53	53
Blackburn	71	73	72	70
QPR	63	63	62	60
Liverpool	59	62	61	59
Sheff. Wed.	59	59	59	59
Tottenham	59	53	52	52
Man. City	57	60	60	59
Arsenal	56	56	58	57
Chelsea	56	54	54	53
Wimbledon	54	57	57	56
Everton	53	55	55	54
Sheff. United	52	57	57	56
Coventry	52	52	53	52
Ipswich	52	52	53	52
Leeds	51	52	53	52
Southampton	50	51	51	51
Oldham	49	50	48	50
Crystal Palace	49	47	47	48
Middlesborough	44	42	41	45
Nottm. Forest	40	40	41	43

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three less than Oldham who finished fourth last.

Sheffield United could have finished with 4 or 5 more points than they did while Aston Villa had 5 or 6 points more than the models predicted. Whilst the Villa conceded only 40 goals, they scored only 57 themselves, just 3 more than Middlesbrough who came second last and were relegated.

According to all criteria, Manchester United clearly deserved to be the champions while the three bottom teams of Crystal Palace, Middlesbrough and Nottingham Forest seemed to be fairly relegated to Division One.

◆ CONCLUSION ◆

Soccer data provide an ideal outlet for statistics students to combine their knowledge of the game and the various tools to come up with alternatives to the current system. Information on matches is readily available from football journals and newspapers.

Other models which might be considered separately include simple ones such as goals scored or goals conceded. However, in the Premier Division these models each had R^2 values of less than 43%.

The conclusion seems to be that the current system of competition points works quite well in most cases but in the event of a tie any one of the three types of models adopted here would be an appropriate discriminator. There is no perfect answer, but one should also be aware of the consequences of using various criteria. For example, using goal averages is going to favour low scoring narrow wins over high scoring narrow wins. That is, a 1-0 victory will be better than a 3-2 win. However, if goal differences are used both scorelines rate the same.

Other models students might investigate include those which take into account home and away performance and other functions of the number of wins, losses and draws. It would also be of interest to determine if the models vary in their appropriateness between divisions.

References

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Table 3. Premier division 1993/94: win/loss/draw record, goals scored for/against and final competition points.

Team	Win	Draw	Loss	Goals for	Goals against	Final points
Manchester United	27	11	4	80	38	92
Blackburn Rovers	25	9	8	63	36	84
Newcastle	23	8	11	82	41	77
Arsenal	18	17	7	53	28	71
Leeds United	18	16	8	65	39	70
Wimbledon	18	11	13	56	53	65
Sheffield Wednesday	16	16	10	76	54	64
Queen's Park Rangers	16	12	14	62	61	60
Liverpool	17	9	16	59	55	60
Aston Villa	15	12	15	46	50	57
Coventry City	14	14	14	43	45	56
Norwich City	12	17	13	65	61	53
West Ham	13	13	16	47	58	52
Chelsea	13	12	17	49	53	51
Tottenham Hotspur	11	12	19	54	59	45
Manchester City	9	18	15	38	49	45
Everton	12	8	22	42	63	44
Southampton	12	7	23	49	66	43
Ipswich Town	9	16	17	35	58	43
Sheffield United	8	18	16	42	60	42
Oldham Athletic	9	13	20	42	68	40
Swindon	5	15	22	47	100	30

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